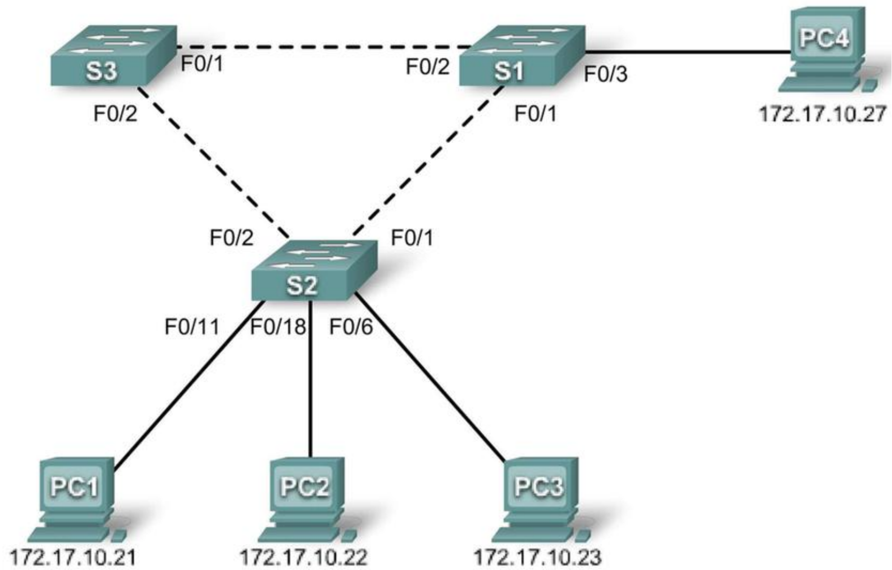
Guia 5– Spanning-tree Protocol

Objectivos:

- Executar a configuração básica de spanning tree nos switchs
- Observar e explicar o comportamento por defeito do Protocolo de Spanning Tree (STP, 802.1D)
- Observar a resposta a uma mudança à topologia de spanning tree



1. Efetue a montagem da rede de acordo com o diagrama anterior. Para implementar os switchs S1, S2 e S3 utilize L3SW.

Device (Hostname)	Interface	IP Address	Subnet Mask	Default Gateway
S1	VLAN 1	172.17.10.1	255.255.255.0	N/A
S2	VLAN 1	172.17.10.2	255.255.255.0	N/A
S3	VLAN 1	172.17.10.3	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.254
PC2	NIC	172.17.10.22	255.255.255.0	172.17.10.254
PC3	NIC	172.17.10.23	255.255.255.0	172.17.10.254
PC4	NIC	172.17.10.27	255.255.255.0	172.17.10.254

2. Configure os PCs de forma a passarem a ter os endereços IP identificados na tabela anterior.

3. Atribua o *hostname* dos switchs de acordo com o nome definido na tabela anterior.

Switch>enable

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```
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z. Switch(config)#hostname S1
S1(config-line)#end
%SYS-5-CONFIG_I: Configured from console by console
S1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

4. Desligue todos os portos usando o comando shutdown.

```
S1(config)#interface range fa0/1-24
S1(config-if-range)#shutdown

S2(config)#interface range fa0/1-24
S2(config-if-range)#shutdown

S3(config)#interface range fa0/1-24
S3(config-if-range)#shutdown
```

5. Religue as interfaces em utilização dos swichs.

```
S1(config)#interface range fa0/1-3
S1(config-if-range)# switchport mode access
S1(config-if-range)#shutdown

S2(config)#interface range fa0/1-3
S1(config-if-range)# switchport mode access
S1(config-if-range)#shutdown

S1(config)#interface range fa0/1-2
S1(config-if-range)# switchport mode access
S1(config-if-range)#shutdown
```

6. Religue as interfaces em utilização dos swichs.

```
S1#show spanning-tree

VLAN0001
```

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```
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0019.068d.6980   This is the MAC address of the root switch

             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0019.068d.6980
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.3    P2p
Fa0/2            Desg FWD 19        128.4    P2p
Fa0/3            Desg FWD 19        128.5    P2p

S2#show spanning-tree

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0019.068d.6980
             Cost        19
             Port        1 (FastEthernet0/1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     001b.0c68.2080
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Root FWD 19        128.1    P2p
Fa0/2            Desg FWD 19        128.2    P2p
Fa0/6            Desg FWD 19        128.6    P2p
Fa0/11           Desg FWD 19        128.11   P2p
Fa0/18           Desg FWD 19        128.18   P2p


S3#show spanning-tree

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0019.068d.6980
             Cost        19
             Port        1 (FastEthernet0/1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     001b.5303.1700
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300


Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Root FWD 19        128.1    P2p
Fa0/2            Altn BLK 19        128.2    P2p
```

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7. Examine a saída dos switchs. O Bridge ID, armazenado no BPDU do spanning tree, consiste na bridge priority, na extensão do system ID, e o endereço MAC. A combinação ou adição da bridge priority e a system ID extension são conhecidas como bridge ID priority. A system ID extension é sempre o número da VLAN. Por exemplo, o system ID extension para a VLAN 100 é 100. Usando o bridge priority padrão de 32768, a prioridade da ID bridge ID priority for VLAN 100 seria 32868 (32768 + 100).

8. Examine a saída dos switchs. O Bridge ID, armazenado no BPDU do spanning tree, consiste na bridge priority, na extensão do system ID, e o endereço MAC. Com base na saída, qual é o bridge ID priority for switches S1, S2, and S3 on VLAN 1?

a. S1 _______
b. S2 _______
c. S3 _______
Qual o root switch para spanning tree da VLAN 1? ________________
No S1, which as portas do spanning tree que estão no estado blocking state switch root? ___________________________________
No S3, which as portas do spanning tree que estão no estado blocking state? ___________________
Como o STP elege o switch root? __________________________
Como as prioridades da bridge são todas iguais, o que mais o switch usa para determinar a raiz? _________________________

9. Observe a alteração da topologia 802.1D STP. Active o debug dos eventos spanning tree.

1. S1#debug spanning-tree events Spanning Tree event debugging is on
2. S2#debug spanning-tree events Spanning Tree event debugging is on
3. S3#debug spanning-tree events Spanning Tree event debugging is on

10. Desligue utilizando o comando shutdown uma das interfaces.
S1(config)#interface fa0/1 S1(config-if)#shutdown

11. Analise a saída do S2 e S3

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Step 3: Record the debug output from S2 and S3

```
S2#
1w2d: STP: VLAN0001 we are the spanning tree root
S2#
1w2d: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1,
changed state to down
1w2d: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to down
S2#
1w2d: STP: VLAN0001 heard root 32769-0019.068d.6980 on Fa0/2
1w2d:     supersedes 32769-001b.0c68.2080
1w2d: STP: VLAN0001 new root is 32769, 0019.068d.6980 on port Fa0/2, cost 38
1w2d: STP: VLAN0001 sent Topology Change Notice on Fa0/2

S3#
1w2d: STP: VLAN0001 heard root 32769-001b.0c68.2080 on Fa0/2
1w2d: STP: VLAN0001 Fa0/2 -> listening
S3#
1w2d: STP: VLAN0001 Topology Change rcvd on Fa0/2
1w2d: STP: VLAN0001 sent Topology Change Notice on Fa0/1
S3#
1w2d: STP: VLAN0001 Fa0/2 -> learning
S3#
1w2d: STP: VLAN0001 sent Topology Change Notice on Fa0/1
1w2d: STP: VLAN0001 Fa0/2 -> forwarding
```

12. Quando o link do S2 que está conectado ao switch raiz cai, qual é a sua conclusão inicial sobre a raiz da árvore de abrangência? ________________________

13. Uma vez que S2 recebe nova informação sobre Fa0 / 2, que nova conclusão ela tira? ______________________

14. A porta Fa0 / 2 no S3 estava anteriormente em um estado de bloqueio antes do link entre S2 e S1 cair. Por quais estados ele passa como resultado da mudança de topologia? _____________________________________

15. Examine o que mudou na topologia da árvore de abrangência usando o comando show spanning-tree

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```
S2#show spanning-tree

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0019.068d.6980
             Cost        38
             Port        2 (FastEthernet0/2)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     001b.0c68.2080
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300


 Interface        Role Sts Cost      Prio.Nbr Type
 ---------------- ---- --- --------- -------- --------------------------------
 Fa0/2            Root FWD 19        128.2    P2p
 Fa0/6            Desg FWD 19        128.6    P2p
 Fa0/11           Desg FWD 19        128.11   P2p
 Fa0/18           Desg FWD 19        128.18   P2p

 S3#show spanning-tree

 VLAN0001
   Spanning tree enabled protocol ieee
   Root ID    Priority    32769
              Address     0019.068d.6980
              Cost        19
              Port        1 (FastEthernet0/1)
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

   Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
              Address     001b.5303.1700
              Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time 300

 Interface        Role Sts Cost      Prio.Nbr Type
 ---------------- ---- --- --------- -------- --------------------------------
 Fa0/1            Root FWD 19        128.1    P2p
 Fa0/2            Desg FWD 19        128.2    P2p
```

16. O que mudou no modo como o S2 encaminha o tráfego? ___________________________________

17. O que mudou na forma como o S3 encaminha o tráfego?
____